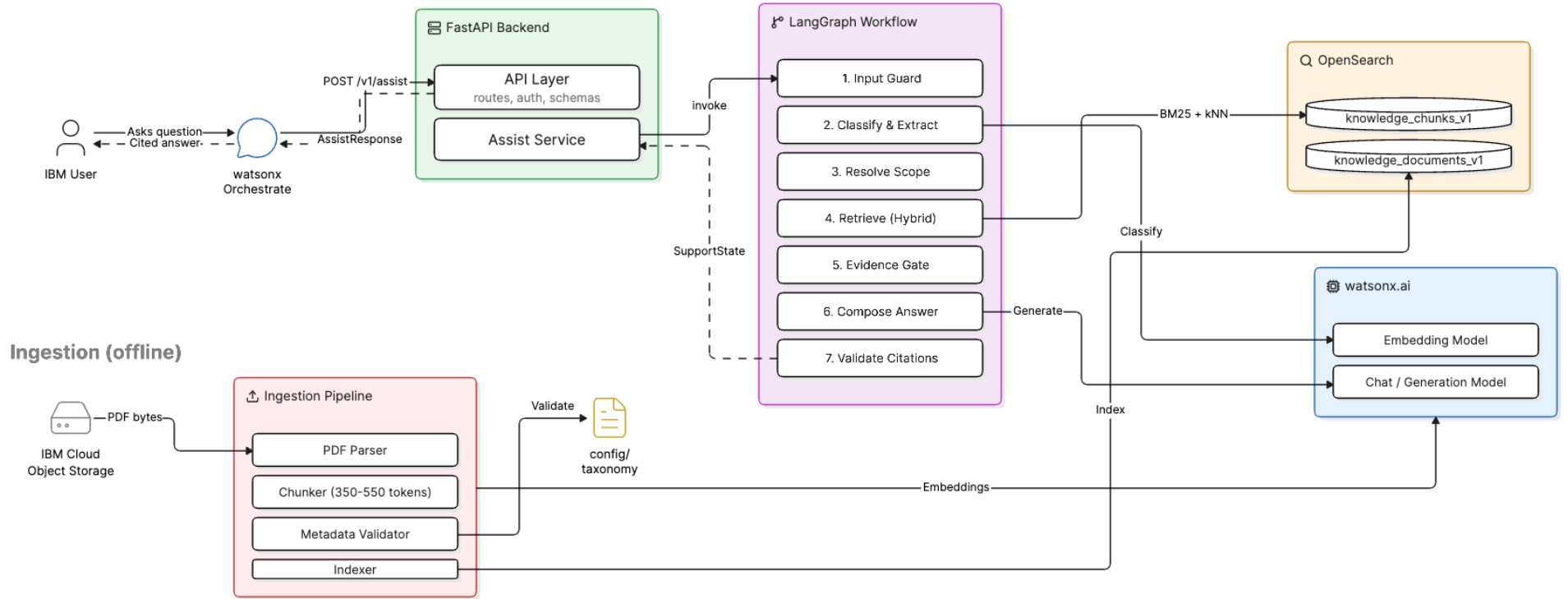


# Weekly Progress Report - Enterprise IT Support Copilot

IBM Internship POC | Status through 10 July 2026 |

## OpenShift & SNO Technical Support Copilot



| Domains                     | Backend             | Retrieval             | Evaluation         | Tests                     |
|-----------------------------|---------------------|-----------------------|--------------------|---------------------------|
| OCP/SNO + Orchestrate + Bob | FastAPI + LangGraph | OpenSearch BM25 + kNN | 38/40 passed = 95% | 58+ focused tests passing |

### Started With

- Built a controlled RAG backend for OpenShift Container Platform and Single Node OpenShift support.
- Created FastAPI service with /v1/assist, /healthz, /readyz, API-key auth, Pydantic schemas, and OpenAPI spec for watsonx Orchestrate.
- Designed a bounded LangGraph workflow for classification, scope resolution, retrieval, evidence checks, answer generation, citation validation, and safety handling.
- Implemented watsonx.ai providers for Slate embeddings and Granite generation.

### Core Delivery

- Built PDF ingestion from IBM Cloud Object Storage/local files using parsing, chunking, metadata validation, embeddings, and OpenSearch indexing.
- Implemented hybrid retrieval using BM25 keyword search, vector kNN search, and Reciprocal Rank Fusion.
- Indexed the approved OCP/SNO corpus, including installation, networking, storage, troubleshooting, authentication, operators, and update guides.
- Added deterministic refusal/clarification behavior for ambiguous, unsupported, out-of-scope, or insufficient-evidence requests.

### Latest Additions

- Expanded the product into a multi-domain Enterprise IT Support Copilot covering OCP/SNO, IBM watsonx Orchestrate, and IBM Bob.
- Added web documentation ingestion through web source discovery plus HTML, Markdown, and plain-text parsing.
- Added new corpus manifests for watsonx Orchestrate and IBM Bob, plus multi-domain taxonomy, classifier, and scope routing updates.
- Fixed non-OCP source labels and Orchestrate payload normalization so IBM Bob/Orchestrate questions route correctly and return product-aware citations.

Current status: working citation-grounded support backend with validated OpenShift/SNO evaluation, multi-domain expansion, Orchestrate integration guidance, Docker/Code Engine deployment docs, and documented next work for broader Bob/Orchestrate evaluation and cross-version retrieval.

- Additionally: The NCR interns have volunteered to support the management for the SKO Event taking part on 13<sup>th</sup> June
- Formed teams for the Watsonx Challeng, brainstorming and formulating ideas for the problem statement